Thanh Tung NGUYEN

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Personal page: tunghawk.github.io

Education

2023 – 2024 **Double Master 2 in Mathematics and Computer Science, Université Gustave Eiffel, France**.
- Awarded Labex - Bézout fellowship.
- Thesis: *Revisit the Sinkhorn barycenter problem* (final result: 16/20).
- Graduated with **high honors**, GPA **14.976/20**.
- Relevant coursework: Data Science basic and advanced (Machine Learning, Deep Neural Networks), Optimization, Statistics, Geometry basic and advanced (graphs), Algorithm and complexity.

2022 – 2023 **Master 2 in Applied Mathematics, Université de Rennes, France**.
- Thesis: *A new numerical approach to stiff ODEs using adaptively optimal time-step*.
- Graduated with **highest honors**, GPA **16.83/20**.
- Relevant coursework: Optimization, Statistics, Bayesian Inference, Differential equations, and Numerical analysis.

2017 – 2021 **Bachelor in Mathematics Teacher, Ho Chi Minh City University of Education, Vietnam**.
- Thesis: *A numerical method to solve the Cauchy problem of 1D Euler fluid dynamic equations*.
- Graduated with **high honors** in major *Analysis and Applied Mathematics*, GPA **8.16/10**.
- Second prize in Mathematics Faculty's student research competition in 2021.

2014-2017 **Vietnamese Scientific baccalaureate specializing in Mathematics, Gia Dinh High School, Vietnam**.
- First prize in city-level Math Contest for 12th-grade students in Ho Chi Minh City.

Interests

Operation research, Optimization, Machine Learning, Optimal Transport, Data Science.

Experiences

03/2025 – 07/2025 **Data Scientist** - New IT Services, Ho Chi Minh City, Vietnam.
- Contribute to construct the Supply Allocation Management (SAM) model that optimizes the supply - demand system: I refined the model by clarifying the objective to maximize profit and enhanced the algorithm by incorporating discount constraints and fixed transportation costs using mixed-integer programming, resulting in more reasonable solution.

04/2024 – 08/2024 **Master 2 Internship** – LIGM, Université Gustave Eiffel, France.
***Topic:** Revisit the Sinkhorn barycenters problem.*
Supervisors: Prof. Théo LACOMBE
Result: We already identified the underlying issue does not come from mistakes either in implementation on *Python Optimal Transport* open-source or in the theoretical results related to the Sinkhorn divergence. It comes from the algorithm's inner workings as the objective function in the optimization problem appears to be flat in some directions when the regularization parameter ε increases, leading to slower or failed convergence. We then achieved a slight improvement in numerical efficiency using accelerated proximal gradient descent. Future work involves optimizing the problem using geometric approaches.

03/2023 – 06/2023 **Master 2 research internship** - INSA Rennes, France.
***Topic:** A New Numerical Approach to Stiff ODEs Using Complementarity Conditions.*
Supervisors: Prof. Mounir Haddou, Dr. Ibtihel Ben Gharbia, Prof. Quang Huy Tran.
Result: Developed a novel variable-based optimization method to adaptively reduce time steps for solving stiff ODEs, resulting in computational efficiency and solution stability. Inspired further research at INSA and IFPEN.

05/2021 – 05/2022 **Mathematics Teacher for high school students** - Titan Education Center, Vietnam.

Experiences (continued)

09/2020 – 04/2021 **Research assistant** - Ho Chi Minh University of Education, Vietnam.
Topic: *A numerical approach to two-dimensional Euler's fluid dynamic equations using the dimensional splitting method.*
Supervisors: Dr. Dao Huy Cuong.
Result: Successfully developed an algorithm combining the Godunov and Upwind schemes to solve the Cauchy problem in fluid dynamics.

Skills

Theoretical	Numerical Optimization, Optimal Transport, Data Science, Numerical Analysis.
Technical	Python (Numpy, Pandas, Scikit-learn, Pytorch Matplotlib), Matlab.
Language	Vietnamese (native), English (proficient), French (beginner).